



# PROJECT-BASED LEARNING

## PHASE-I EVALUATION

**Disaster Resource Allocation and  
Shelter Management System**

Team ID: OSDBMS-T135

**Department of Computer Science & Engineering**  
Graphic Era (Deemed to be University), Dehradun  
Academic Session 2026–27

## Team & Mentor Details

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### Team Information

**Team ID:** T135  
**Semester:** 5th  
**Section:** ML1,B,E,C  
**Domain:** OSDBMS

### Team Members

1. *Anushreya Tomar – 24022202*
2. *Akanchha Singh – 24012753*
3. *Ishani Nautiyal – 24022264*
4. *Simarjeet Kaur – 24021248*

### Mentor

*Dr. Ankit Tomar*

**Interactions completed: 3**

# Problem Statement & Motivation

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## Problem Statement

*During emergencies like floods and earthquakes, shelter registration is done manually using paper, phone calls, or WhatsApp to manage resource distribution. This may lead to shelter overcapacity, overdue resource allocation and problems in identifying urgent aid cases. The project looks at the problem of lack of centralized and real-time system for registration of shelters and distribution of resources.*

## Motivation

- ▶ *Manual and fragmented management*
- ▶ *Uneven resource and shelter allocation*
- ▶ *Lack of real-time information*

## Target Users / Stakeholders

- ▶ *Relief workers, shelter administrators*
- ▶ *Disaster-affected people and local govt. organizations*
- ▶ *Faster, fairer and more organized allocation*

# Objectives

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## Project Objectives

1. *Create a system to control shelters and observe their occupancy.*
2. *Implement a priority system to provide assistance to people in urgent needs such as injured persons, elderly people, pregnant women.*
3. *Provide a real-time monitoring of shelters and available resources through a dashboard.*
4. *Provide allocation in a concurrency-safe way to avoid overbooking or double booking.*
5. *Provide automated triggering of alerts and redirection if a shelter reaches 90 or already full.*

**Key Objective:** *Provide a centralized, real-time and reliable system for efficient shelter and resource allocation during disasters.*

## Proposed Approach

1. *Gather data regarding registrations, capacity of shelters, requirements in every location.*
2. *Implement a centralized processing system and manage the allocation of shelters and resources.*
3. *Prioritize the urgent cases like injured, old-age citizens.*
4. *Maintain the database of shelters, registrations, transaction records.*
5. *Monitor resources through a dashboard, and redirect people to the nearest available shelter.*

## Key Features

- ▶ *Real-time shelter resource monitoring*
- ▶ *Priority-based assistance*
- ▶ *Concurrency-safe allocation*
- ▶ *90 capacity alerts*
- ▶ *Automatic nearest-shelter redirection*

# Integration of PBL Course Concepts

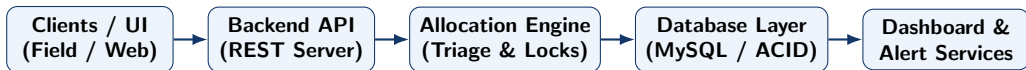
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| Course                   | Concepts Applied                                                  | Project Module                       |
|--------------------------|-------------------------------------------------------------------|--------------------------------------|
| <b>Operating Systems</b> | <i>Concurrency, Synchronization, Concurrent Requests</i>          | <i>Concurrency-Safe Allocation</i>   |
| <b>DBMS</b>              | <i>Tables, Transactions, Data Management, Concurrency Control</i> | <i>Shelter and Resource Database</i> |

**Requirement:** Apply OS concepts for concurrency-safe allocation and DBMS concepts for reliable storage and transaction management.

# System Architecture / Workflow

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## Major Components

- ▶ **Intake & Auth API:** Ingests evacuee priority flags and manages volunteer auth.
- ▶ **Triage & Allocation Logic:** Ranks arrivals by urgency and reserves beds atomically.
- ▶ **Alert & Redirection Engine:** Dispatches 90% capacity warnings and computes nearest routes.

## Data / Control Flow

- ▶ Intake forms submit triage attributes to the backend REST API.
- ▶ Transactions lock capacity records during allocation to avoid race conditions.
- ▶ Live dashboard updates occupancy metrics; overflow triggers automatic redirection.

# Technology Stack & Methodology

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## Technology Stack

- ▶ Programming: *Python*
- ▶ Database: *MySQL*
- ▶ Tools / Framework: *Web-Based Dashboard*
- ▶ IDE: *VS Code*
- ▶ Version Control: *Git / GitHub*

## Development Methodology

1. Requirement Analysis
2. System Architecture Design
3. Database and Backend Implementation
4. Allocation and Concurrency Implementation
5. Testing and Evaluation

A centralized web-based architecture with database storage, priority-based allocation, concurrency control, real-time monitoring, and automated alerts is used to provide reliable and efficient disaster shelter management.

# Initial Progress & Team Contribution

## Progress Achieved

- ▶ *Problem and requirements identified.*
- ▶ *Project goals and milestones defined.*
- ▶ *System Architecture and workflow planned.*
- ▶ *Database and resource-management approach defined.*
- ▶ *Core allocation, concurrency, and alert mechanisms planned.*

## Individual Contribution

| Member           | Role                           | Contribution                                                                        |
|------------------|--------------------------------|-------------------------------------------------------------------------------------|
| Anushreya Tomar  | Database + Concurrency         | MySQL database, tables, transactions, concurrency, synchronization, locking         |
| Akanchha Singh   | Frontend + Dashboard           | Web interface, registration form, shelter dashboard, occupancy and resource display |
| Ishani Nau-tiyal | Allocation + Priority Logic    | Triage, priority-based allocation, shelter assignment, capacity reservation         |
| Simarjeet Kaur   | Alerts + Redirection + Testing | 90% alerts, nearest-shelter redirection, testing, load testing, documentation       |

# Roadmap, Expected Outcomes & References

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## Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

## Expected Outcomes

- ▶ *Functional web-based disaster shelter management system.*
- ▶ *Real-time shelter occupancy and resource dashboard.*
- ▶ *Priority-based allocation for urgent cases.*
- ▶ *Load-testing simulator for 50+*

## References

1. Altay, N. & Green, W. G. (2006). OR/MS Research in Disaster Operations Management. *EJOR*, 175(1).
2. Vanajakumari, M., Kumar, S. A., et al. (2012). Humanitarian Logistics and Disaster Management. *IEOM*.
3. Caunhye, A. M., Nie, X., & Pokharel, S. (2012). Optimization Models in Emergency Logistics. *SEPS*, 46(1).

**Thank You**

Questions & Suggestions